MIT 2.003 SC PS1 solving

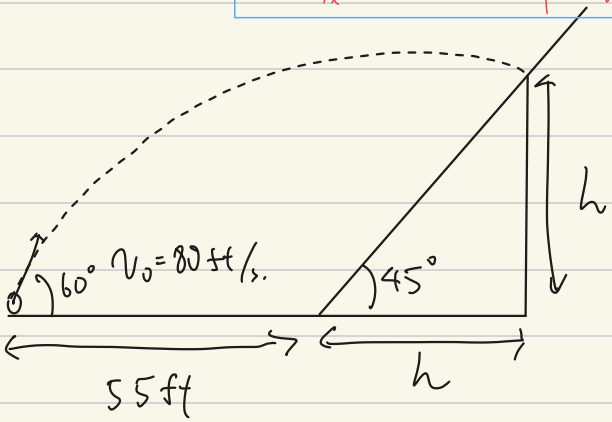
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Made with Goodnotes

Problem 1:

Notation disclaimer:

$\vec{v}_{A/X}$: velocity of A wrt X | $\hat{i}, \hat{j}, \hat{k}$ etc are directions, unit vectors of x, y, z resp.



$\downarrow j$

$1m \approx 0.308ft$

$9.8 ms^{-2} \approx 31.9 ft s^{-2}$

Find h:

Step 1: Assuming ball will travel further than 55ft which is for sure. And assuming if hits h ft in to the stands, or lands with a vertical displacement of h ft.

Step 2: Use the horizontal displacement as a way to express time. Substituting that into the results of step 1:

Step 1: Using $s = ut + \frac{1}{2}at^2$

$h = v_0 \sin 60° t - \frac{1}{2} 31.9 t^2$

$h = 40\sqrt{3}t - 15.95t^2$

Step 2:

$v_0 \cos 60° t = 55 + h$

$t = \frac{55+h}{40}$

Results:

$h = 40\sqrt{3}\frac{55+h}{40} - 15.95\left(\frac{55+h}{40}\right)^2$

$1600h = 1600\sqrt{3}(55+h) - 15.95(h^2 + 110h + 3025)$

$0 = -15.95h^2 + (1600\sqrt{3} - 3354.5)h - 48248.75 + 88000\sqrt{3}$

$h \approx -101.1$ rejected or ≈ 64.6 ft (using quadratic formula)

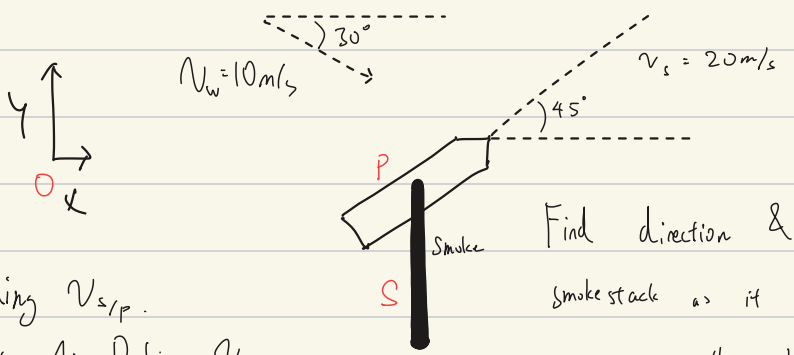
Total horizontal displacement:

$64.6 + 55$

≈ 119.6 ft

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Problem 2:



Find direction & magnitude of smoke stack as it appears to a passenger on the ship.

Finding $v_{s/p}$.

Step 1: Define $v_{w/o}$

Step 2: Observe what $v_{s/p}$ means.

Step 3: find the answer

Step 1: $v_{w/o} = v_{w/p} + v_{p/o}$

Step 2: Smoke stack comes out at rest, the only force acting upon it is the wind, and the smoke travels with the wind at the same velocity of the wind itself.

Hence our answer is simply the $v_{w/p}$ itself.

$v_{w/p} = v_{w/o} - v_{p/o}$

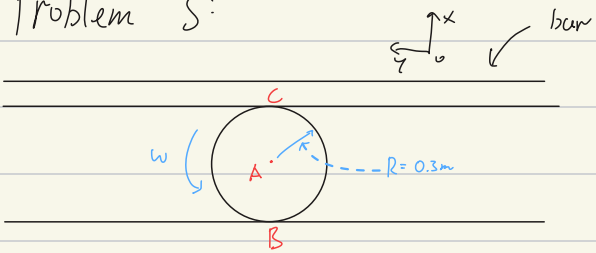
$v_{w/o} = 10\cos 30^\circ \hat{i} - 10\sin 30^\circ \hat{j}$

$v_{p/o} = 20\cos 45^\circ \hat{i} + 20\sin 45^\circ \hat{j}$

$v_{w/p} = (5\sqrt{3} - 10\sqrt{2})\hat{i} + (-5 - 10\sqrt{2})\hat{j}$

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Problem 3:



Notation: $r_{A/B}$ is position of A wrt B

Pre-req knowledge.

$$v_{A/B} = \left(\frac{\partial r_{A/B}}{\partial t}\right)_{w=0} + \vec{\omega} \times r_{A/B}$$

(no rotation)

Find velocity of bar at contact C pt.

Step 1: Identify ICR

Step 2: Express $v_{C/O}$

Step 3: Find $v_{C/O}$

Step 1: ICR is B.

Step 2:

$$v_{C/O} = v_{B/O} + v_{C/B}$$

$$v_{C/O} = v_{B/O} + \left(\frac{\partial r_{C/B}}{\partial t}\right)_{w=0} + \vec{\omega} \times r_{C/B}$$

$\left(\frac{\partial r_{C/B}}{\partial t}\right)_{w=0}$ is 0

$$v_{C/O} = v_{B/O} + \vec{\omega} \times r_{C/B}$$

At that instantaneous time,

$v_{B/O}$ is 0

$$v_{C/O} = \vec{\omega} \times r_{C/B}$$

$$v_{C/O} = \vec{\omega} \times r_{C/B}$$

$$= 4\hat{k} \times 2(0.3)\hat{i}$$

$$= 2.4\hat{j}$$

Answer:

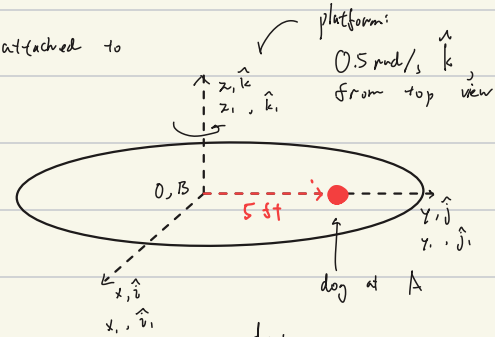
$v_{C/O} = 2.4 \text{ ms}^{-1}\,\hat{j}$

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Problem 4:

B_{x_1,y_1,z_1} is attached to the platform

$Oxyz$ is fixed



In this moment of time x_1, x and y_1, y are aligned!

dog: + 2ft/s radially outwards along positive y_1 axis

$v_{A/B} = 2\text{ft/s}\,\hat{j}_1$

Find velocity of dog wrt fixed inertial frame $Oxyz$.

Step 1: Express $v_{A/O}$

Step 2: Solve intermediate velocities to obtain answer.

Step 1:

$$v_{A/O} = v_{B/O} + v_{A/B}$$

Step 2:

$$v_{A/O} = v_{B/O} + \left(\frac{\partial r_{A/B}}{\partial t}\right)_{\omega=0} + \vec{\omega} \times r_{A/B}$$

$$= 0 + 2\text{ft/s}\,\hat{j} + 0.5\hat{k} \times 5\hat{j}$$

$$= 2\hat{j} - 2.5\hat{i}$$

Pre-req:

Although $v_{A/B} = 2\text{ft/s}\,\hat{j}_1$, $\hat{j}_1$ is not static, hence $v_{A/B}$ expressed in $\hat{i}, \hat{j}$, must require:

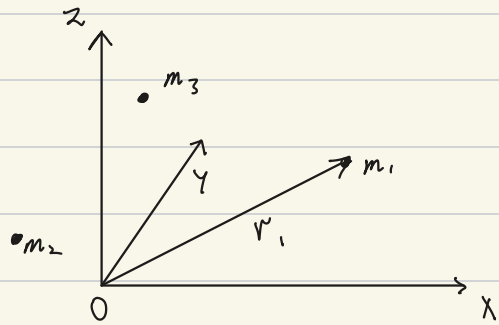
$$\left(\frac{\partial r_{A/B}}{\partial t}\right)_{\omega=0} + \vec{\omega} \times r_{A/B}$$

Answer:

$$v_{A/O} = 2\text{ft/s}\,\hat{j} - 2.5\text{ ft/s}\,\hat{i}$$

Made with Goodnotes

Problem 5:



$m_1 = 3m$

$m_2 = m$

$m_3 = 5m$

where $m = 1kg$

$m_1(2, 2, 5)$ $V_1 = (1, -4, 3)\,V_0$

$m_2(-2, 4, 3)$ $V_2 = (1, 0, 0)\,V_0$

$m_3(4, 0, 2)$ $V_3 = (-2, -1, -1)\,V_0$

where $V_0 = 10 m/s$

a: find COM

b: Compute P_{total} of three

c: Compute $V_{G/0}$, COM velocity wrt O

d: Compute T_{total}, KE_{total} of all three particles

a: Finding COM: (COM: where total mass is concentrated)

using weighted avg method:

$$(m_1 \cdot m_1\,pos + m_2 \cdot m_2\,pos + m_3 \cdot m_3\,pos) / (m_1 + m_2 + m_3)$$

$$= (3(2, 2, 5) + (-2, 4, 3) + 5(4, 0, 2)) / (9)$$

$$= \left(\frac{8}{3}, \frac{10}{9}, \frac{28}{9}\right)$$

b: Finding P_{total} of system

$$\vec{P} = \sum_i m_i V_{i/0} = 3(1, -4, 3)10 + (1, 0, 0)10 + 5(-2, -1, -1)10$$

$$= (-60, -170, 40)\ \frac{kg\,m}{s}$$

$$= (-60\hat{i} - 170\hat{j} + 40\hat{k})\ \frac{kg\,m}{s}$$

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c: computing $v_{G/O}$

$$= \frac{\sum_i m_i v_{i/O}}{\sum_i m_i} = \left| \frac{(-60\hat{i} - 170\hat{j} + 40\hat{k})}{3+1+5} \right|$$

$$= \left| -\frac{20}{3}\hat{i} - \frac{170}{9}\hat{j} + \frac{40}{9}\hat{k} \right| \approx 20.52 \text{ m/s}$$

d: Find T_{total}

$$T_{total} = \frac{1}{2}\sum_i m_i v_{i/O} \cdot v_{i/O}$$

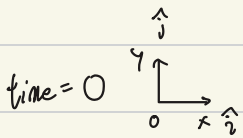
$$= \frac{1}{2}(3(1+16+9))10^2 + \frac{1}{2}10^2 + \frac{1}{2}(5(4+1+1))\times 10^2$$

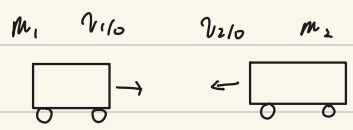
$$= (39 + \frac{1}{2} + 15) \times 10^2$$

$$= 5450 \text{ J}.$$

Problem 6

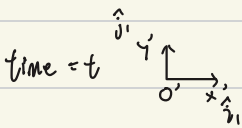
observing from fixed inertial frame $Oxyz$





$m_1 = 1000\text{kg}$ $m_2 = 2000\text{kg}$

$v_{1/O} = 25\text{m/s}\,\hat{i}$ $v_{2/O} = -25\text{m/s}\,\hat{i}$





$|v_{1/O}| = |v'_{1/O'}|$ only possible if $v_{O'/O}$ is non-zero

a: Find change in P_{total} & change in T_{total} (KE)

b: You are riding a train, // to road, $v_{train/O} = v_{1/O}$,

Repeat a) calculation, state general obs.

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a: Find change in P_{total} & T_{total}

Finding $v'_{1/0}$: $v_{1/0} m_1 + v_{2/0} m_2 = (m_1+m_2) v'_{1/0}$

$$v'_{1/0} = -\frac{25}{3} \text{ ms}^{-1} \hat{i}$$

$P_{total\ before} = \sum_i m_i v_{i/0} = m_1 v_{1/0} + m_2 v_{2/0} = 1000 \times 25 - 2000 \times 25$

$= -25000 \frac{kgm}{s} \hat{i}$

$P_{total\ after} = \sum_i m_i v_{i/0} = (m_1+m_2)\left(-\frac{25}{3}\right) = -25000 \frac{kgm}{s} \hat{i}$

$P_{total\ before} = P_{total\ after}$, 0 change of P_{total}.

T_{total} change:

$T_{total\ after} - T_{total\ before} = \frac{1}{2}(m_1+m_2) v'_{1/0} \cdot v'_{1/0} - \frac{1}{2} m_1 v_{1/0} \cdot v_{1/0} + \frac{1}{2} m_2 v_{2/0} \cdot v_{2/0}$

$= \frac{312500}{3} - 812500 + 625000$

$= -8.33 \times 10^5$ J

b: We are now computing wrt O', where

$v_{O'/0} = 25 \text{ m/s} \hat{i}$, $v_{1/O'} = 0$, $v'_{1/0} = -\frac{25}{3}$

$v'_{1/0} = v'_{1/O'} + v_{O'/0}$

$v'_{1/O'} = -\frac{100}{3} \text{ m/s} \hat{i}$

$v_{2/0} = v_{2/O'} + v_{O'/0}$

$v_{2/O'} = -50 \text{ m/s} \hat{i}$

$P_{total\ change} = P_{total\ after} - P_{total\ before} = (m_1+m_2) v'_{1/O'} - m_2 v_{2/O'}$

$= -100000 + 2000 \times 50$

$= 0$

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$P_{total\ change} = 0$

- Finding $T_{total\ change\ wrt\ O'}$

$$T_{total\ after} - T_{total\ before} = \frac{1}{2}(m_1+m_2)v'^2_{1/O'} \cdot v'^2_{O/O'} - \frac{1}{2}m_1 v^2_{1/O'} \cdot v^2_{1/O'} + \frac{1}{2}m_2 v^2_{2/O'} \cdot v^2_{2/O'}$$

$$= \frac{1}{2}3000\left(\frac{100}{3}\right)^2 - \frac{1}{2}1000(0)^2 + \frac{1}{2}2000(50)^2$$

$$= 1666666.667 - 2500000$$

$$= -\frac{2500000}{3}$$

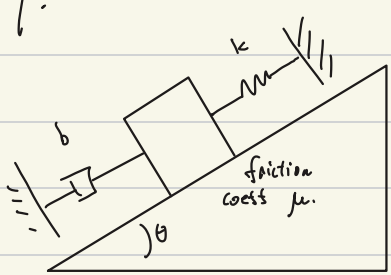
$$\approx -8.33 \times 10^5\ J$$

- Conclusion:

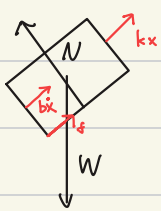
$\Delta T_{inertial} = \Delta T_{O'}$ The conservation of energy and

$\Delta P_{inertial} = \Delta P_{O'}$ linear momentum is conserved for all inertial reference frames (no acceleration)

Problem 7:



- a) fbd.



a: fbd

b: Assign coordinates

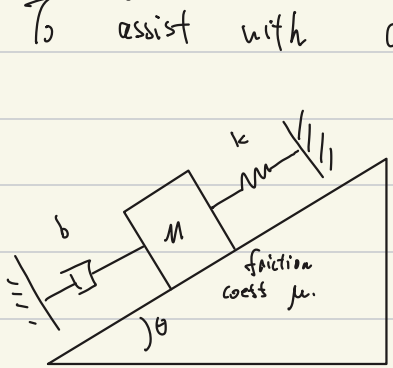
c: EOM for mass on incline.

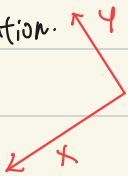
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b) Assigning coordinates.

There are many arbitrary answers that are all right.

To assist with calculation.





where $x = 0$ for 0 spring force at that instantaneous moment.

c) Finding EOM.

Assuming tendency of motion

Using $\sum_i F_i = ma$ in positive direction.

Assuming $x=0$ has 0 spring tension.

$$mg\sin\theta - b\dot{x} - kx - \mu mg\cos\theta = m\ddot{x}$$

$$m\ddot{x} + b\dot{x} + \mu mg\cos\theta + kx = mg\sin\theta$$

Answer

$$m\ddot{x} + b\dot{x} + kx + \mu mg\cos\theta = mg\sin\theta$$

Made with Goodnotes